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Roll No

EC-4005-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Control Systems

Time : Three Hours

Maximum Marks: 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) State the effects of adding poles to $G(s)H(s)$ on the root locus.
b) Determine the stability of a system whose characteristic equation is $s^4 + 5s^3 + 20s^2 + 40s + 50 = 0$.
2. a) With suitable example explain phase lag, phase lead and phase lag-lead compensator in time domain.
b) Discuss about derivative and integral compensation.
3. a) What is the need of compensators? Draw the diagram of lead compensator and find the expression for its transfer function.
b) Explain the working of PID controller and its advantages.
4. a) What are the limitations of open loop system over closed loop system?
b) What are the characteristics of feedback?

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5. a) A feedback system has a closed loop transfer function as

$$\frac{10(s+4)}{s(s+1)(s+2)}$$

Construct state model and its representation

- b) Write a short notes on various steps for constructing the Bode Plot and mention its advantages.
6. a) Discuss about all pass and minimum phase systems.
b) Discuss Nyquist stability criterion.
7. Sketch the root locus for the open loop transfer function of a unity feedback control system given as

$$G(s) = \frac{k}{s(s+1)(s+3)}$$

Determine the value of k for damping ratio of 0.5 and for marginal stability.

OR

- a) What is meant by PI control? State the effect of PI controller on the system performance.
b) Derive expression for the transfer function of a phase lead compensator.
8. Write short notes on (any four)
- a) Laplace Transform
 - b) Effect of location of poles on stability
 - c) Minimum-Phase systems
 - d) Gain Margin and Phase Margin
 - e) Transfer Matrix

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